AUTOMATION TEST RESULTS

Project: Factorial Web Application
Tester: Percival Gebashe
Date: 19/03/2026
Framework: Playwright (Java) + TestNG

1. Test Execution Summary

Metric	Value
Total Tests	24
Passed	16
Failed	8
Skipped	0

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2. Test Results Breakdown

Functional Tests

TC-01 → Pass

TC-02 → Pass

TC-03 → Pass

TC-04 → Pass

Boundary Tests

TC-05 → Fail (Infinity issue)

TC-06 → Fail (500 error)

Validation Tests

TC-07 → Fail (no backend validation)

TC-08 → Pass

TC-09 → Pass

TC-10 → Pass

TC-11 → Pass

TC-12 → Pass

UI / UX Tests

TC-13 → Pass

TC-14 → Pass

Navigation Tests

TC-15 → Pass

TC-17 → Fail (placeholder content)

TC-18 → Fail (placeholder content)

API Tests

TC-18 → Pass

TC-19 → Pass

TC-20 → Fail (UI/API inconsistency)

TC-21 → Fail

TC-22 → Fail

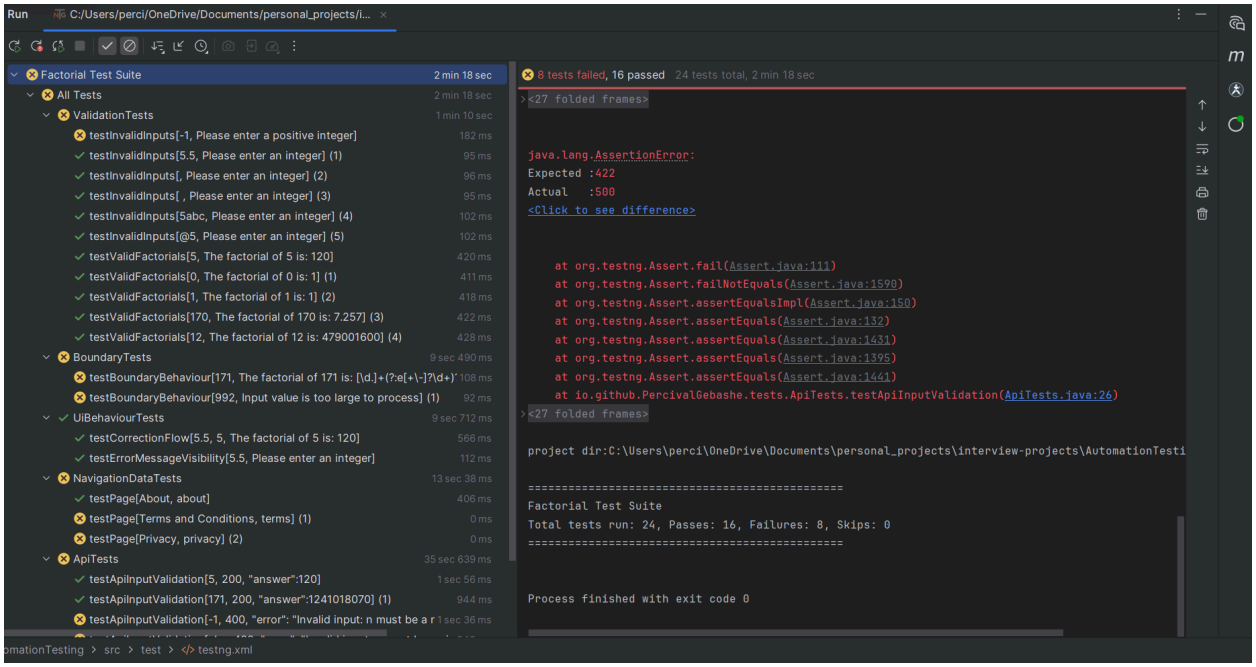
TC-23 → Fail

Additional Tests

TC-24 → Fail

3. Evidence (Screenshots)

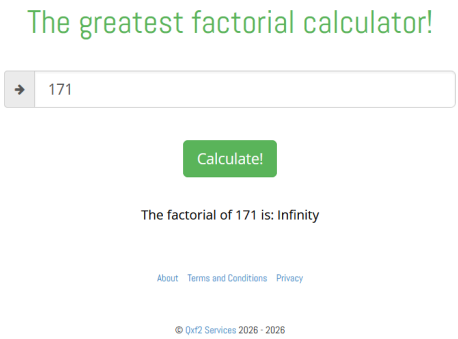
Screenshot 1: Successful Test Execution



TestNG console output

Screenshot 2: Failed Boundary Case

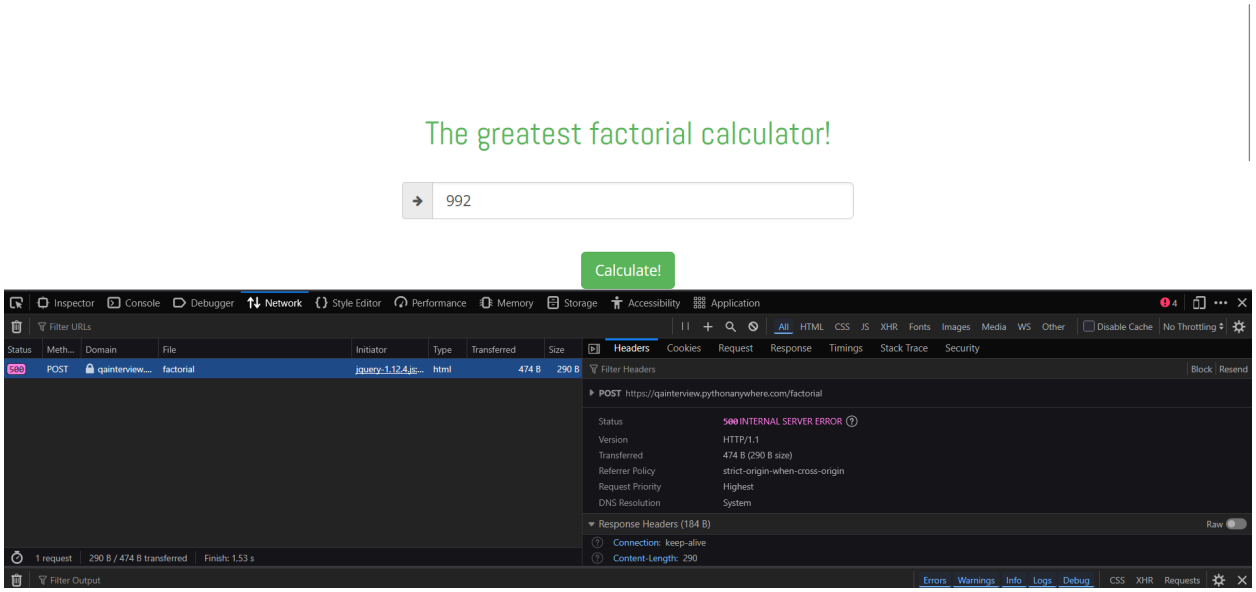
Input: 171



Output: Infinity

Screenshot 3: Server Error

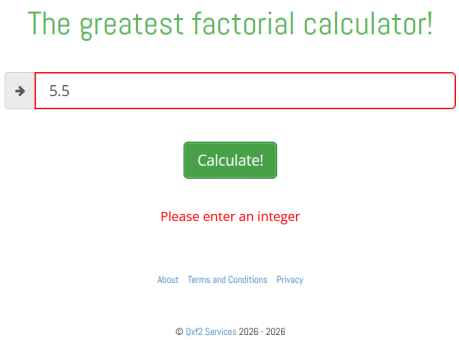
Input: 992



Output: 500 error

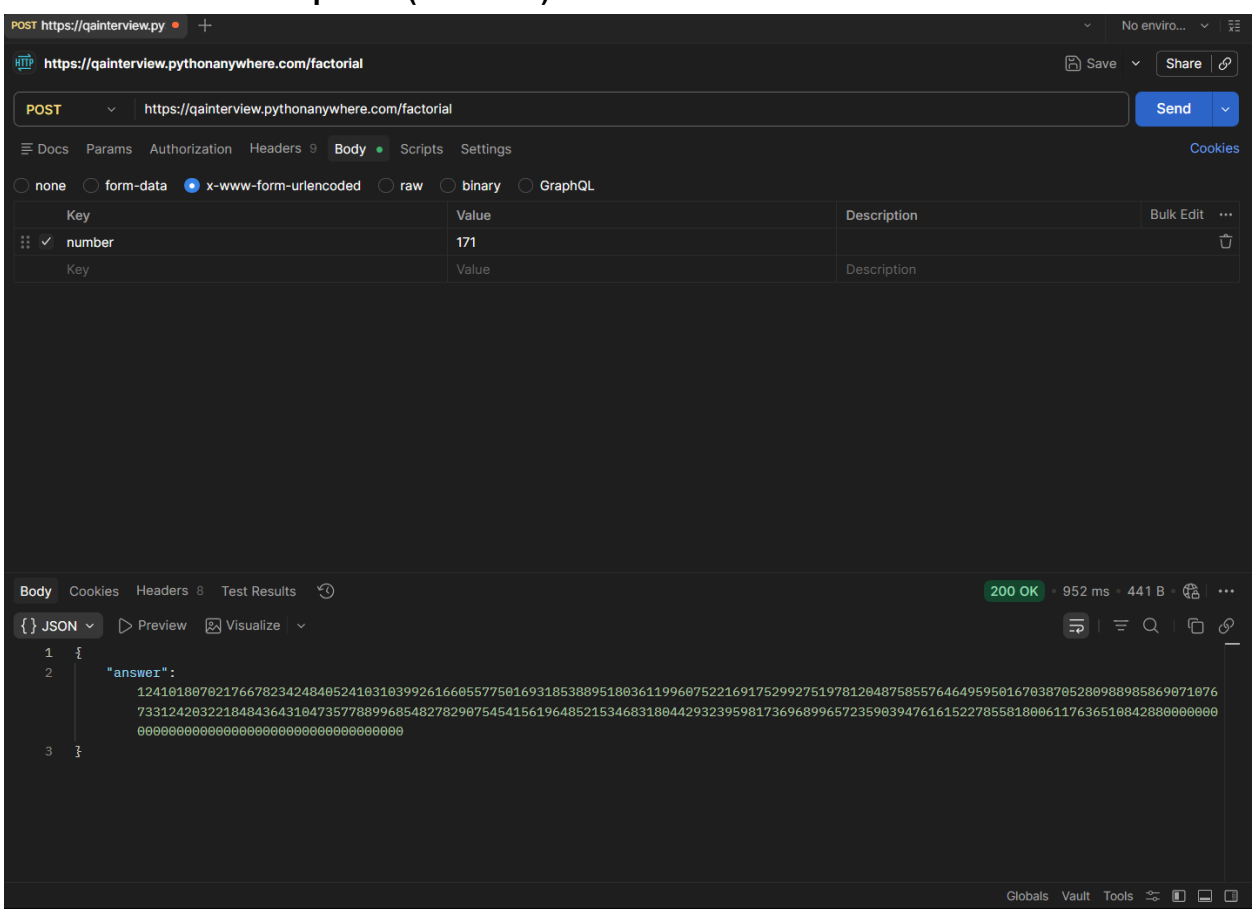
Screenshot 4: Validation UI

Invalid input (5.5)



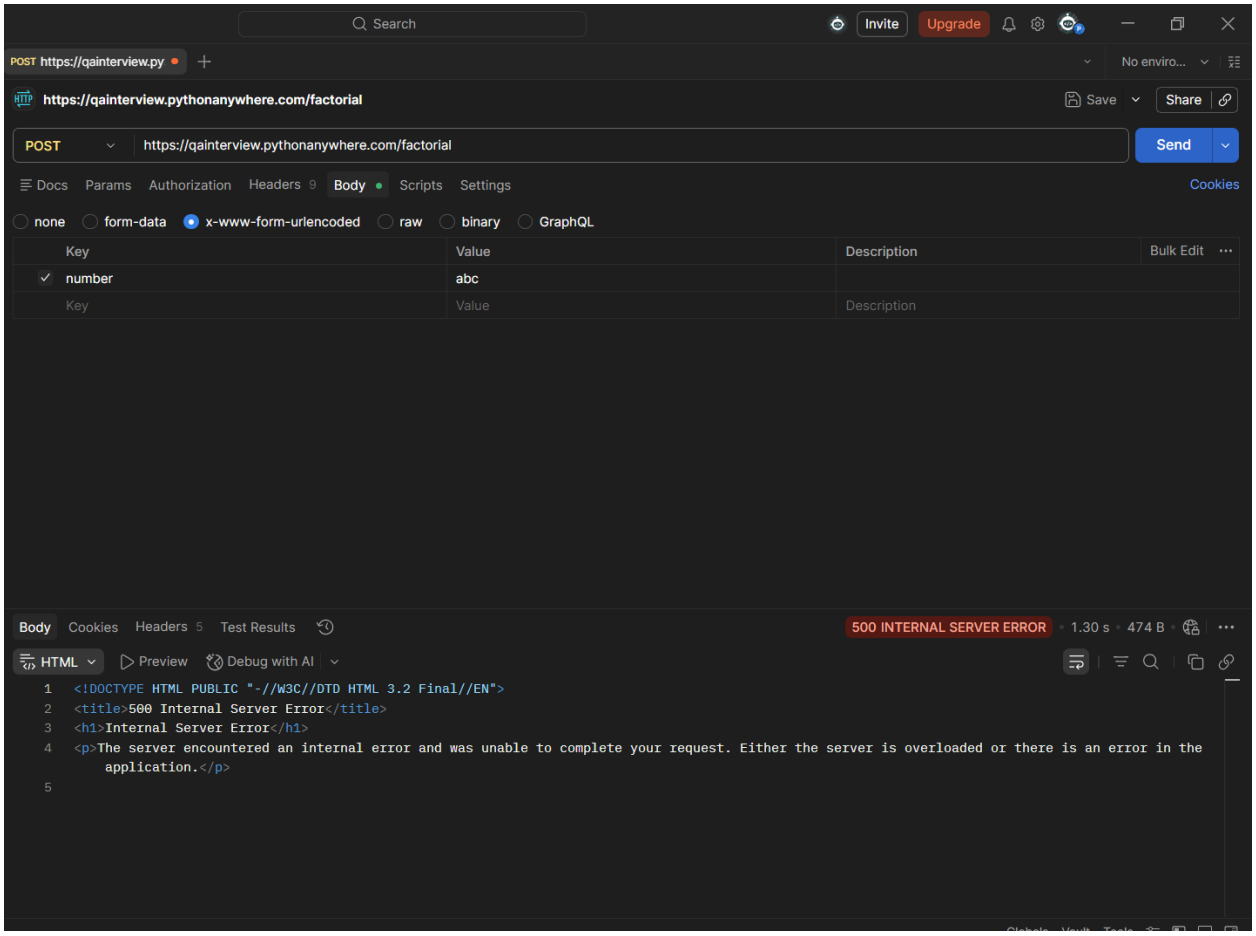
Error message + red highlight

Screenshot 5: API Response (Postman)


POST https://qainterview.py
No enviro...
https://qainterview.pythonanywhere.com/factorial
Save
Share
POST
https://qainterview.pythonanywhere.com/factorial
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
number
171
Key
Value
Description
Body
Cookies
Headers 8
Test Results
200 OK
952 ms
441 B
JSON
Preview
Visualize
{
"answer":
1241018070217667823424840524103103992616605577501693185388951803611996075221691752992751978120487585576464959501670387052809889858690710767331242032218484364310473577889968548278290754541561964852153468318044293239598173696899657235903947616152278558180061176365108428800000000000

number=171 → full factorial

Screenshot 6: API Error Case



number=abc → 500 error

4. Key Observations

Backend supports large number calculations accurately

UI fails to represent large values correctly

Backend lacks input validation (returns 500 instead of controlled errors)

System behaviour is inconsistent across layers

5. Conclusion

The automation suite confirms that while the application performs correctly for standard inputs, it lacks robustness in handling edge cases, validation, and consistency between frontend and backend behaviour.